

Interpretable Deep Learning for Bank Churn Retention

Capstone Project · Team 11

Баяндинова Амина · Золотухина Евгения · Мурчич Наталья

Predict churn probability, rank customers by risk, and turn the MLP output into a targeted retention workflow.

0.8574

ROC-AUC

3.93x

top-decile lift

27.6%

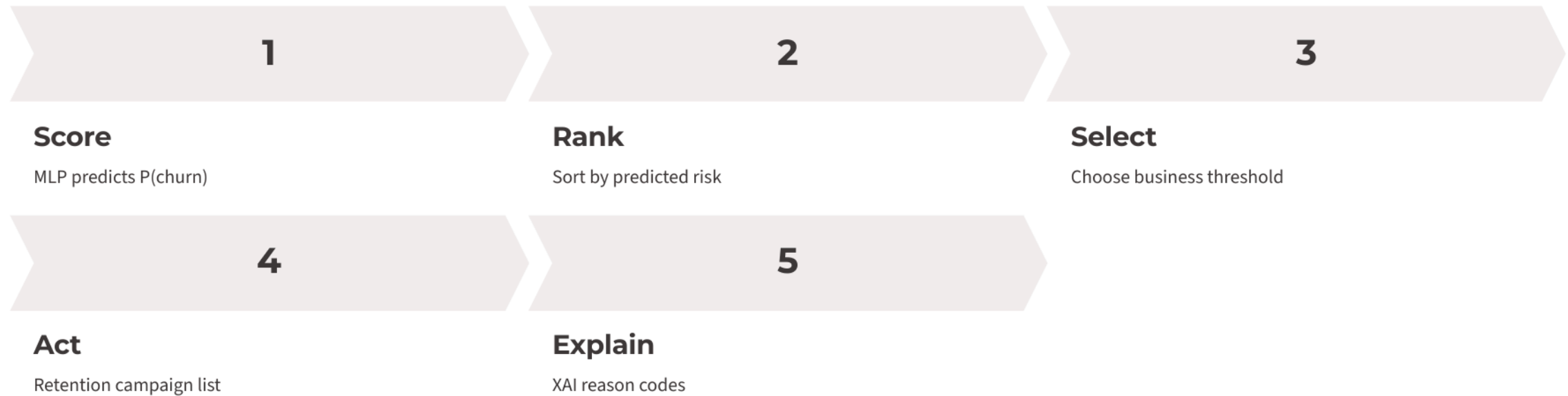
contact rate



Business problem: proactive retention

The model answers who is likely to churn, how risky they are, and why

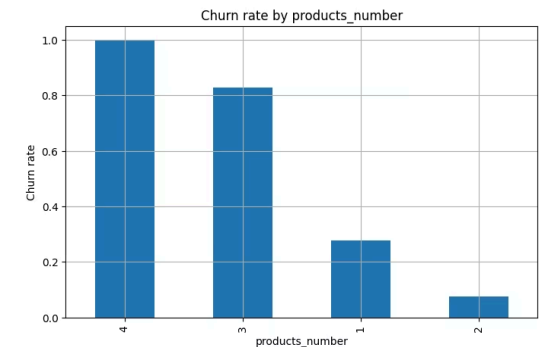
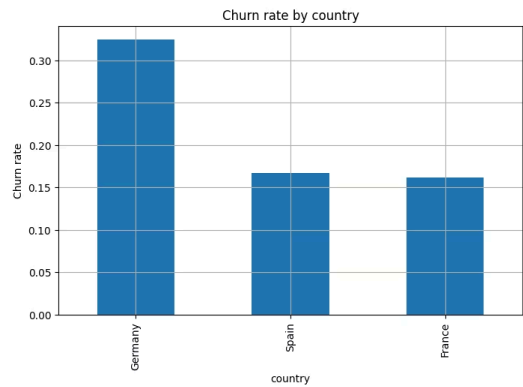
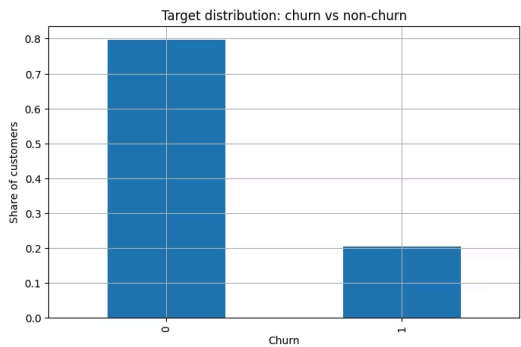
Banks cannot contact every customer. The business goal is to rank customers by churn probability and select a manageable high-risk group for retention actions.



Output: a prioritized retention list, not a broad campaign to the whole customer base.

Data and feature space

10,000 customer-level records · no missing values · target churn $\approx$ 20%

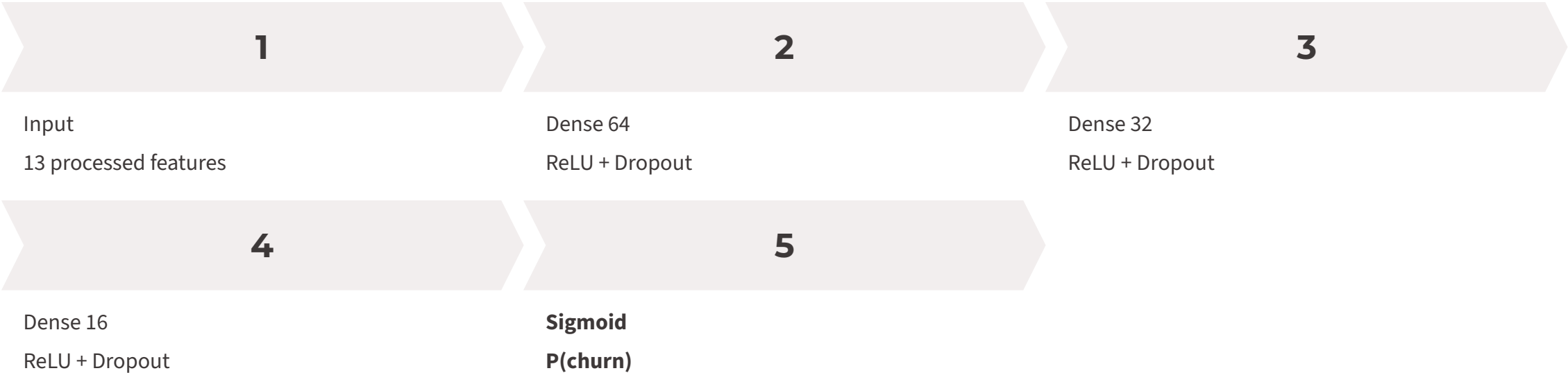


Feature group	Examples	Business meaning
Demographics	age, gender, country	customer segment and market context
Financial profile	credit_score, balance, estimated_salary	capacity and relationship value
Product usage	products_number, credit_card	depth of relationship with the bank
Engagement	active_member, tenure	activity and loyalty signals

customer_id is excluded as a technical identifier. EDA shows churn differences by activity, product count, age and balance; with only ~20% churn, accuracy alone is not an informative KPI.

Modeling approach: deep MLP for tabular churn

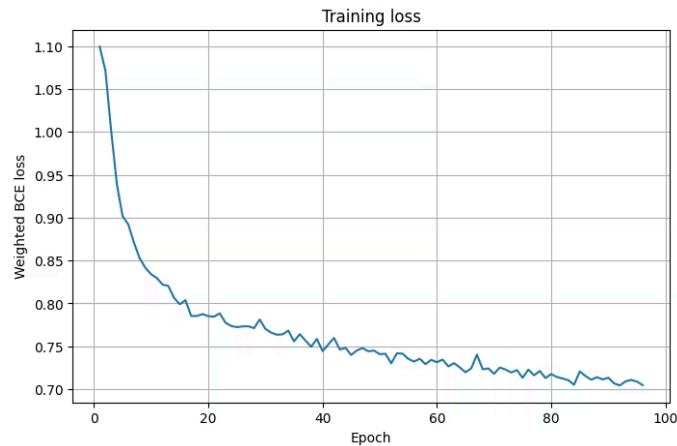
Dense layers learn nonlinear interactions between customer attributes



Training design choice	Why it is used
Weighted binary cross-entropy	Churn is minority class; false negatives are costly
Dropout regularization	Reduce overfitting on tabular data
Early stopping	Stop when validation ROC-AUC no longer improves
Threshold optimization	Choose an operating point aligned with retention economics

Training and threshold selection

Validation ROC-AUC stabilized around 0.858; selected threshold = 0.57



0.57

selected threshold

3.908

positive class weight

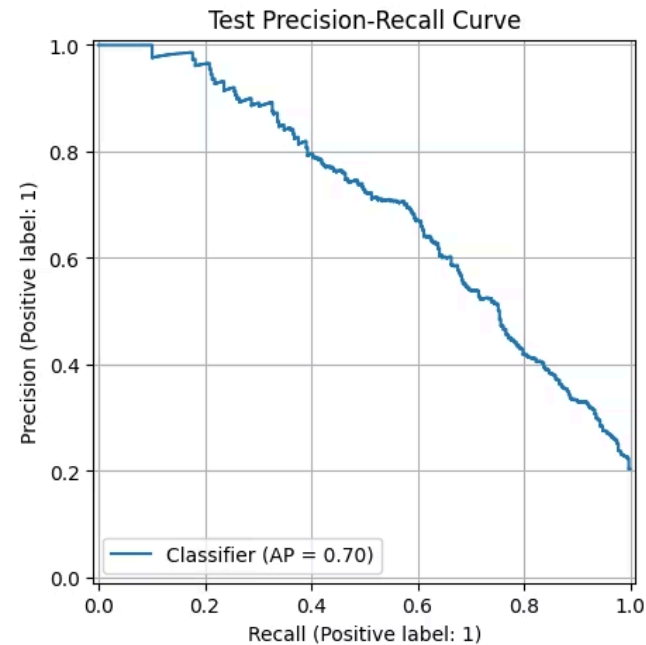
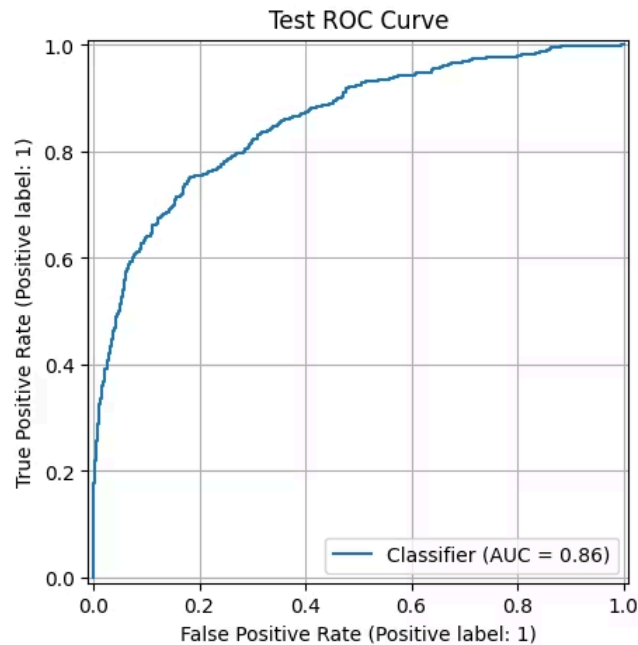
05

early stopping epoch

The default 0.50 cutoff was not used. The threshold was chosen on validation data using business assumptions: capture many churners while keeping contact volume manageable.

Holdout performance

Strong ranking quality and positive-class performance



0.8574

Test ROC-AUC

0.6975

Test PR-AUC

0.6090

F1-score

ROC-AUC is the primary metric because the bank needs a reliable risk ranking, not just a 0/1 prediction.

Operational result at selected threshold

The model contacts 27.6% of customers and captures 71.7% of churners

	Predicted stayed	Predicted churned
Actual stayed	1,333	260
Actual churned	115	292

292

true churners targeted

260

false positives contacted

115

churners missed

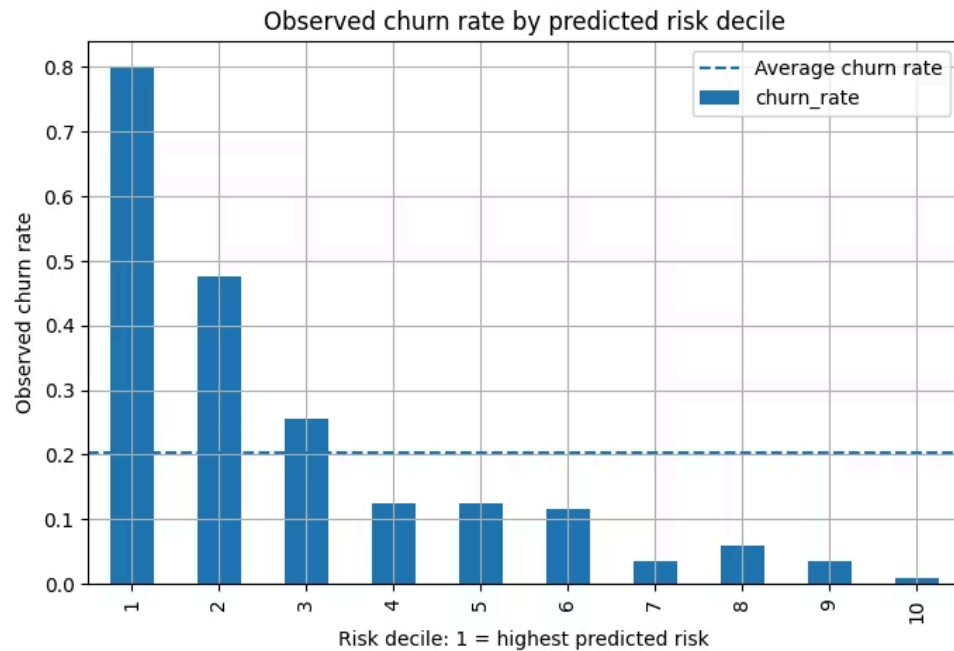
552

customers contacted

Business interpretation: false negatives are lost retention opportunities; false positives create campaign cost. The operating point favors recall because missing a churner is expensive.

Lift analysis: the strongest business result

Top 10% of predicted-risk customers have 80.0% observed churn



80.0%

churn rate in top decile

20.35%

average test churn rate

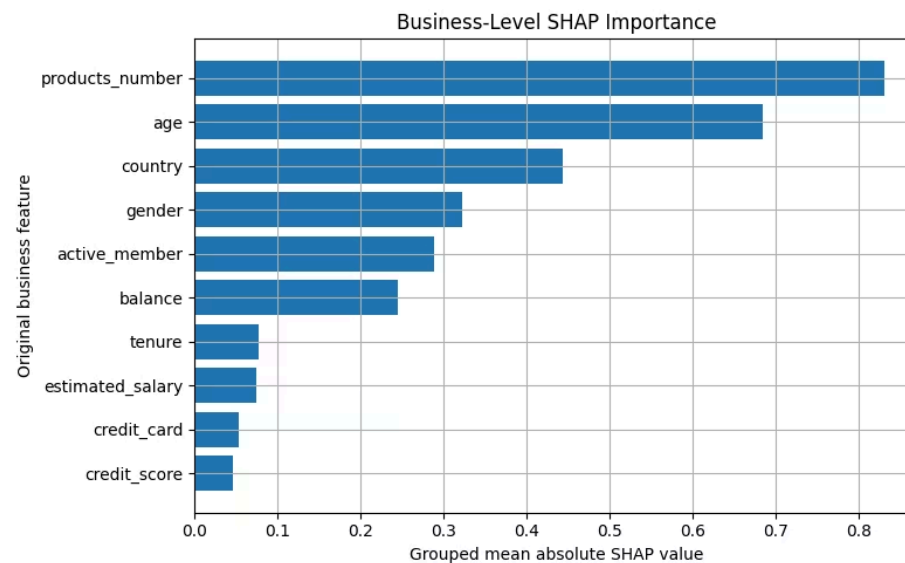
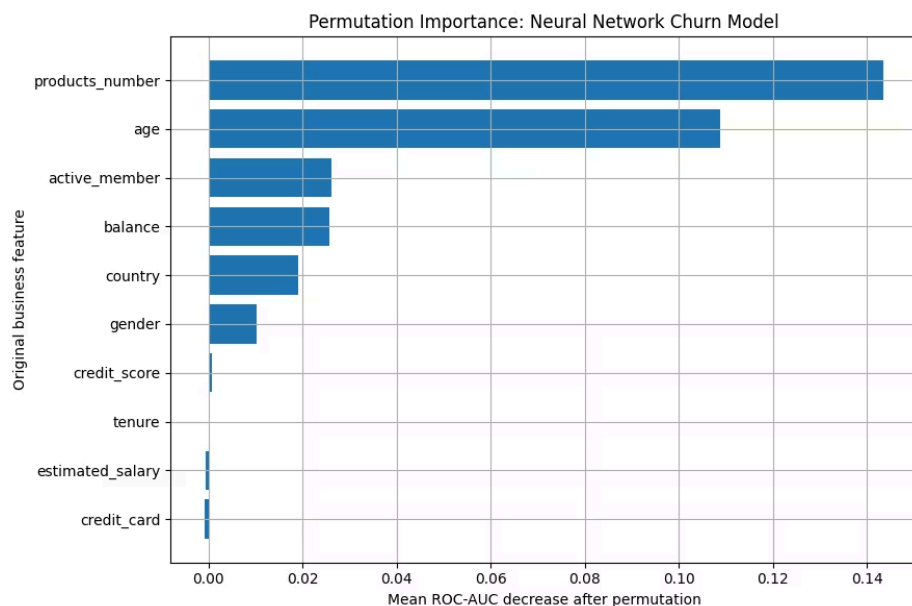
3.93x

top-decile lift

Retention strategy suggested by the notebook: top 10% get high-priority personalized actions; next 10–20% get lighter offers or proactive communication; low-risk groups are monitored.

Explainability: global churn drivers

Permutation importance and SHAP identify plausible business signals



Global explanations: permutation importance and SHAP identify products_number, age, active_member, balance and country as the strongest drivers.

Local explanations: SHAP and LIME turn an individual risk score into customer-level reason codes. SOM analysis additionally visualizes risk regions in the MLP latent space.

Business impact simulation

Illustrative economics connect model output to retention value

Impact metric	Value
Customers contacted	552
True churners targeted	292
False positives contacted	260
Expected saved customers	10
Expected retained value	36,500
Campaign cost	27,600
Expected net impact	8,900

Formula used in the notebook: expected saved value = true churners contacted × retention success rate × customer value; net impact = saved value – campaign cost.

8,900

expected net impact



Important limitation: the financial simulation is illustrative and must be calibrated with real bank economics before production deployment.

Conclusion

A predictive model plus XAI makes churn management operationally usable

The project delivers an interpretable churn-management workflow: customer data → preprocessing → MLP churn probability → threshold/ranking → retention list → XAI explanation → targeted action.

Strong ranking

ROC-AUC 0.8574 and PR-AUC 0.6975

Business focus

Top decile lift 3.93x; contact only 27.6%

Explainability

Permutation, SHAP, LIME and SOM support trust

Actionability

Outputs a prioritized retention list and impact simulation